

What is Active Inference?...with Daniel Friedman

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method: auto_caption

hello and welcome to the innocence channel is there a way to talk about how we perceive the world act in the world and make sense of the world that also explains how other organisms communities and systems operate right from ant colonies to teams of work colleagues today we have daniel friedman helping us understand a conceptual framework called active inference which may be able to do just that daniel has a fascinating academic background in biology and genetics his phd was studying ant behavior he co-founded the active inference lab i really hope you enjoy this conversation and can see how sensing inner bodily sensations or interception fits with this active inference please remember to subscribe to the innosense channel for more information about the role of your bodily sensations and tips for using bodily awareness to improve your well-being thank you okay so good morning daniel i believe it's pretty early on in your homeland yes yes it's it is early in california 6am or 7am okay so thanks ever so much for taking the time to or making the time so early on to join us on the inner sense channel um we're here today to talk about a topic called active inference so please could you tell me a little bit about your background and how you ended up establishing the active inference lab which is something that i've really enjoyed viewing it's an online participatory lab that's communicating learning and practicing applied active inference well thanks for the invitation and yes looking forward to unpacking a lot of this and i'll just begin with my path and how we all together came to working in this active inference lab so i've been a student of biology and genetics always very curious about how different living systems worked and enacted in their environments and i went to graduate school did a phd in biology and specifically that was in behavioral ecology in ants and so for five years in graduate school i did field work in southern arizona in the united states and also a lot of lab work and computational work and all those experiences are kind of embodied in their own way and help just show me some glimmers of how many different methods and approaches and perspectives there were on just one question like how do ants make decisions about how to regulate their foraging behavior at the colony level so that's kind of what i've been interested in from a

research perspective and during that time i was also learning more about different frameworks for perception cognition and action and there are many such frameworks but the one that i started to learn more about in around 2017 and 2018 was the free energy principle and active inference so mm-hmm there i'll just leave there before we go to the next chapter yes wow so i'm i'm just curious about ants and um how you study them in the field so literally you were out these weren't ants that had been collected in a kind of glass container these were in a literal field were they yes they were in an actual open grassy field and ant field work was awesome the methods ranged from highly improvised basic like using toothpicks and spray bottles of water and stop watches and little flags and a lot of just things that are available objects and questions that people could ask in their local neighborhood or in their backyard counting ants crossing one direction or another on the trail but then also there was the liquid nitrogen and collecting ants and labeling them and giving them pharmacological treatments and then collecting them and measuring different aspects of their gene expression and all that comes with that in terms of like the molecular biology and the computation so ants are awesome to study and i hope that we can continue to tie that in in this conversation because although your wall and some of your previous discussions focus on the human sometimes i just think about things in terms of the ants and the colony and um i think that just shows how when we do have integrative ways to talk about perception cognition and everybody will have their own systems that they've studied and phenomena that they've looked into but there will also be like a higher language that helps us connect the dots across those different systems okay yeah and i think that word language as you just said is um sometimes it's uh key i think you had um was it mark solms on your channel a while back and he um said this phrase i think it was a avoid who's wanting to avoid getting into semantic tangles and i think that is so such a you know i love that that phrase stayed with me and it's so um such a useful thing to remember when trying to have these sorts of discussions that as you said we can be coming from different perspectives and have different experiences and different languages to describe what we think is a useful way to explain our existence but one person's choice of words might have a very different semantic uh reasoning or understanding to someone else's so um yes let's proceed um one thing i wanted to mention now that the theme tune for the active inference lab is excellent was there any particular reason why you chose that theme tune was it did it represent the free energy principle or active inference or it's got some stochastic elements and some nuns the theme song was a special request of a friend of one of the co-founders of the lab yvonne and uh dimashvakov is the name of the musician at least i did not provide even any suggestions

but when i hear it i hear how there's a regimentation and an orderliness but also an energetic feeling and a motivational and maybe even an action-oriented feeling to it so it always is exciting when i hear it um let me return to the story of the lab so in uh 2019 i connected with the other people who we would all together go on to co-found the lab and that was alex and yvonne and we were very interested in applying active inference to organizational design especially to online teams and organizations they had a background in systems engineering and systems thinking i had a background in thinking about complex systems like the ants and also learning about active inference and altogether we knew that we had a lot of learning and applying ahead of us and so we thought how will we rise or meet this unknown unknown and only parts of it we were in a position to see at that moment how will we meet this large challenge of applying active inference a framework which is still undergoing conceptual development just like all conceptual frameworks are but also one that we believe has a lot of possibilities for application and so we thought we're not in a position to be hiring people to be in an academic lab getting graduate students and the pace on that would be slow even if that were an affordance that we had an opportunity for action and so we wanted to take the spirit of open science decentralized science and just participatory sense making and bring some of these threads together by offering this co-creative space the active inference lab where participants from different backgrounds different time zones different levels of familiarity with active inference and other topics can come together and be learning communicating and applying active inference that's uh really nice to hear the background to that and the kind of the the spirit of it being so open and open to people joining in and i would encourage anyone who's interested after they've heard what active inference is to to subscribe to your channel and you always take time in the first of a series of digesting papers to kind of really slow things down and to um cover all the the different concepts that you're going to talk about in a particular series so let's start by just um in a nutshell can you explain what is active inference please okay well the first thing that comes to mind with a question about nutshells is hamlet who said i could be bounded in a nutshell and count myself a king of infinite space active inference is very austere in some ways because the kernel of this nut the action perception loop very pre-application let's say it's very general and it needs a lot of meat to be added and more to that skeleton before there's something there to simulate or use even as a thinking tool so the kernel of the nut is simple almost to the point of being infuriatingly so like language so that would be my zeroth pass would be the nutshell is its own domain and inside of it there's so much space to be developing and there's so much that will sprout out of it but yes good

okay so the infuriatingly simple um bits about action and perception what is that so if there's someone you've never never spoken to before you meet them um in an elevator what would you say action active inferences okay so here's how i would uh perhaps approach that it's all in the title active inference is about integrating action and and then if we go one level deeper we're using inference which is to say like cognition or in a computer happens to be computation we're using inference as a way of integrating our models and perspectives on with cognitive processes that include planning perception emotion memory anticipation so we have this tremendous diversity of sensory processes those could be our visual sense our olfactory sense our interoception proprioception so we have many inputs to the system and then there's also a diversity of actions that different systems select and those can be like the sort of on-board action opportunities the on-board affordances for bodily action and also they can include extended affordances like using a computer or using a pencil so we have a lot of inputs and then we have a lot of outputs so a lot of sense coming in a lot of actions coming out and then there's a lot of things that happen in between in what we call broadly cognition again like some of those features or phenomena i mentioned like planning anticipation memory etc now we could have bespoke narrow theories for each sensory modality for each cognitive phenomena and for each action affordance and or we might be interested in pursuing some integrative ways to unify our formal models and our informal models of perception cognition in action so i believe that is what active inference offers and then we can go into in a second how it differs from some other frameworks but i believe what it offers is a way to think about that input space the cognitive space and the output space as well as in the impact which is kind of the fourth quadrant there thinking about all of those in a unified way which at the very least we have to do for useful computational modeling but also just for useful sense making so it really unifies all of the components of our being you'd say of our how we experience ourselves is that going too far each will use the map to navigate the territory so if it's honest in how you say it then amazing i hope so how does it sorry i was just going to say how does it differ then from you know other more classical ways of viewing how living organisms go about their business of staying alive awesome question i hope or as we would say an active expect and prefer that in the future there will be a url that i can provide where these will be listed and enumerated and more referenced but just off the cuff how does active inference differ from other potentially integrative theories for action perception and cognition whether they be cybernetic or more system specific like a human psychological theory so one key difference between active inference and some frameworks for biological and computational behavior like

reinforcement learning is that in active inference action selection is accomplished by always the joint consideration of epistemic and pragmatic value epistemic value means the value of information or of learning and knowledge like epistemology and pragmatic value is like utility or reward and so in reinforcement learning there's an emphasis on the common currency of pragmatics reward value and then it leaves a bit of this awkward or ad hoc way that curiosity and intrinsic motivation has to re-enter and be translated or transmuted into this common currency of pragmatics like a curiosity bonus right we'll pay you a little bit to go on an extra epistemic foraging trip in active inference though we can have a common grounding between epistemic and pragmatic value which enables our simulations and our embodied systems to rapidly make decisions that in one moment are more exploratory and in the next moment can be more exploitative over multiple time scales first point is it's a way that different than reinforcement learning we can bring epistemic and pragmatic value into a common ground the second short and that's as mentioned earlier that in contrast with essentially any other framework although i'm open to hearing what else people bring to the table it integrates very diverse and heterogeneous perceptual cognitive and action features in at least an inter-accommodating framework and so very few other theories or frameworks attempt to have such a scope for example there might be a model of just memory or of just action selection but then those are like different modules that then have an ad hoc connection and so active inference gives a principled set of connections and then the last point that i'll make in terms of differentiation with active inference versus many other theories is with active inference we can be quite explicit about where are we talking about the map and where are we talking about the and so the territory how things quote really are and the map how we're modeling it and so i think about a linear regression and if we did a linear regression between height and weight and we found that there was a positive or negative correlation or however it may be nobody would think well you're saying that height and weight are a linear regression because the linear regression is just a map it's just a model of a territory and so similarly with active inference though many times people are interested in how things quote really are how the territory is and may even use active inference to discuss as if they are discovering how the territory is we also have this opportunity to be very clear about map making and i think that is a tremendous value because it helps us hold a space between the observations that we're taking from a system and the modeling that we're doing while also leaving some of the mystery and openness okay some of the mystery and openness um the use of you know words terrain and map mate making there i think um will be useful later on when we dive into some of the concepts uh of that are

used amongst your colleagues and peers in the lab um especially when it comes to this idea of you know sometimes it appears like the terrain and the map are identical you know day to day as i go about my daily life i feel that what i you know things happen so quickly that it feels like um i'm actually that the terrain and the map equal each other but what i'm hearing here saying here is that um gives us an opportunity to understand how the map is formed and how it relates to the changing terrain is that right so we're all without falling into any simple fallacies were all inside of our sensorium pick a different spatial metaphor if you choose but we are not directly observing lightning strikes in clouds we're receiving sensory input in the form of sound and light in that case of like lightning and thunder and then we're doing inference on a hidden cause an unobserved cause again though we observe the sensory consequences of the territory and we can take a totally action oriented realist pragmatic let's get out of this open field because there's a thunder and lightning storm happening so we can make useful action while also holding that space and not thinking well we won't be able to take useful action until we're in that cloud directly observing the lightning strike it's like no we are observing the light and the sound and we're making an inference about unobserved causes and we can share that with each other and we can decide how to act based upon that and then maybe when we're hanging out in the house we could talk about what is real and what lightning really might be but there's also this ability to be applying active inference again at the level of mapping while also holding space for some philosophical questions about the nature of the territory i think this is getting to a really key point here so the sensorium um way i understand it is that that's just that's a way of describing the the collection of different things we might sense so it's our a bit like um you know we have the genome and the microbiome or collections of genes and microbes but the sensorium is our global collection of sensors so not just the five sensors as you said um the uh interception proprioception all of these things um maybe ph blood pressure temperature uh changes in air pressure outside of us is anything that we might sense either consciously or subconsciously yep and if i could bring in one term from a field that active inference draws a lot on which is ecological psychology um ecological psychology provided that notion of affordance which is like a capacity for action so it's something that could be selected on the outbound and then a sort of complement to the term affordance is umvelt and yes it's a german term or something like that and i'm sure it could be looked up and searched more but it is exactly referring to that sensorium which is that integrated perceptual experience and inactive inference that is actually an actively generated process and we can experience that directly with our visual input so in the anatomy of the eye there's actually color receptive cells

that are more concentrated in the center of the eye so there's higher resolution and there's color vision in the center of the eye and that actually kind of drops off in color perception and in uh clarity as it goes out and of course there's a quite large blind spot in the eye which is like where the optic nerve is coming out yet our visual experience or at least a possible visual experience has the felt sense of color in all parts of the visual field equal clarity and quite curiously no blind spot right so even just in the visual modality before we get to multi-sensory integration or some of these more ineffable senses we're already quite aware that that Umwelt that sensorium is not just a signal processing it's not just the microphone and then the filter and then the central processing unit and then the action selection unit there's actually something quite dynamic where even when we close our eyes we can still continue to have a generative model but when our eyes are open that too is a generative model one that is meeting a compromise with the incoming sensory information to be sure but also is actively generated by the generative model of the organism just to sort of quote i think it's Anil Seth from his book Being You he's he says that all of our perceptions are controlled hallucinations that happen um with through because of you know our body so this so it's not just visualizing things we visualize with our eyes closed that are kind of uh models actually everything that we perceive constructed as a kind of uh to help a model to help us understand things which are beyond our direct ability to ability to and to perceive directly yes i think we can even explore this notion of perception as active or generative in two senses so the first is in this sense of the generative model for example a vision where there's this um generated visual scene what Anil Seth calls the controlled hallucination that is being met with a compromise by incoming sensory data and that can be modeled using Bayesian techniques but we won't go down that road so that's one sense that perception is generative and active just to summarize is it's not this passive receiving it's something endogenous internal and then being met with a compromise with internal uh incoming information and the other sense that perception is quite importantly active it's an amazing example that's used in the recently introduced active inference textbook and i think it also speaks to this map territory distinction and the importance of action for perception and so the experiment the embodied experiment goes like this place one's finger on their pants and so that is i'm gonna do this now so on my thigh just when you say pants in English sorry in England means underpants but i can do it on anywhere on my trousers now i have reinterpreted many of my previous awkward interactions now i'm just kidding yes on one's favorite piece of clothing or fabric on the couch if they may is the fabric rough and the answer is actually to even determine that the finger must be moved because the roughness is a

is a feature that we do active inference on through actual physical motion by reducing our uncertainty through we're integrating action and perception and so we couldn't just sit there so to speak with our finger unmoving and think about whether it's rough or not so here the fabric is like the territory and the map is how rough is it there maybe there's parts where it's rough and there's parts where it's smooth but how could we know if all we have is a finger in a dark room we need to move it and that actually has a lot to do with other cases of active perception like ocular motor movement eye motion when we're reading or scanning or all other kinds of things that people can think of where the sensory inputs are not predestined or determined but they're actively sought after in a way that is more like seeking and informed foraging rather than just ad hoc sampling that's a really simple but powerful examples so i can't tell if something's rough or smooth um [Music] i could guess it if i was looking at it but by touch i have to move i have to take this active part of the to infer whether it's rough or smooth and so that's a kind of an easy to grasp physical example because um we all know how to move our fingers and feel texture um but what we're saying is that that principle applies to all types of inference um yeah it's act there's an active component and i guess that makes sense with sound because we can only you know it's the moving between pictures isn't it it's the change that allows us to interpret sound if it was all just one tone and constant we would never be able to distinguish between anything audible and and just one other example so we've mentioned the physical palpation with touch and touch receptors and so it's a yes and with all the biology of the touch receptors so this is something that is here to again be a yes and to all the molecular biology of touch it's just adding in this note that in the organismal context action is part of the perceptive process then we mentioned the visual scanning and then a third example might be in communication where somebody might have uncertainty about whether they're conversant on the same page as them or seeing what they're seeing or whether they're grasping or comprehending so it's like we already have these visual and tactile and even prehensile or grasping metaphors for communication and cognition and we might take an active step of like asking how does that sound to you or what does that make you think about or does that make sense what we're planning here those are an example of communication as action and inference in the interpersonal and that's an example about how active inference can be applied at the cellular and at the organismal level but also at the interpersonal level which again speaks to one of the important aspects of this integrative framework which is that it helps us connect the dots with different systems and skills and um that helps me appreciate the the concept of an affordance as well which you you mentioned you can't really you know so an affordance

from what i understand is that the quality of something so you know a cup um has got grasp ability i can hold it but to um you know to a fork laying on the table that the the cup isn't graspable because the fork isn't doesn't ever use cups so it's that the fact that we have to actively do something that makes the cup have the affordance of being graspable and an important piece there is that relationality or the way that you described how the affordance it's not like we could just tag the cup as being grabbable by whom not even by all humans and so actually the concept of an affordance and this radically relational nature of affordances and situational helps us bring accessibility into the picture bring personalization and just other situational context because there's no such thing as an affordance that is agnostic to the actor again it's an example of a qualitative insight like interactions are relational or preferences expectations affordances all are conditional on the situation those are quite qualitative felt senses that many people may resonate with they might not have on their own connected that to bayesian statistics and formal models of the action perception loop however we're all increasingly finding that indeed those connections are there and hopefully we can hold the space for there to be a continuum of discussion again between those qualitative insights like context matters and some formal models for action and perception in context okay um can we uh think about i'm looking at your mission statement again for the active inference lab so it's um a participatory online lab that's communicating learning and practicing applied active inference can um we think about the practicing applied part so in what ways can active inference be applied once you become aware of the concept how can you apply it or practice it awesome well i look forward to a thousand applications blooming so i have more uncertainty than certainty on that question because i think we're just beginning a few areas where active inference has been applied if i could list them and then we can see which ones might be interesting the first is in the area of bodily physiology whether in terraception as well as in mental health and in cognitive dynamics so there's a lot of work that's been done in a clinical and in a psychological a second area is in the group level so thinking about online teams and communities as we and others have and modeling group communication as a case of active inference and then the third might be in robotics and embodied non-biological systems or systems that we want to imbue with some natural perceptive cognitive and active features but are not the same type of biological thing as we are so those would be three quite interesting areas would be the organismal and bodily the community team group communication space and the embodied robotics i find that the so uh the second one the hardest to to kind of see how that be applied i can see how you know in as you said mental health and um yeah in the

health domain it's it's really easy to apply or possible to apply and also robotics or artificial intelligence because you know it's all very much about being able to take action and infer and all of that but for the the group dynamics um now can you give a sort of example about how the concept of active inference applies in a group setting i'm thinking about your ants in the field now yes yes is the ant the nest mate or is the ant the colony well aren't colonies just sets of nest mates but aren't nest mates just sets of organs and aren't organs just sets of cells and so on and how do we think about collective action again somebody may have no issue with thinking about an organism as being the locus of perception cognition and action people make decisions right groups also make decisions sometimes explicitly like through what is quite literally called policy decision making in governance or in an organization we also call that policy selection in active inference so sometimes groups make overt policy selection and there's some type of internal process of determining that like a vote or however it's done so just as we could think about the subunits whether they're modeled or unmodeled in the organism leading to an organismal decision about policy we could also think about an analogous process be it implicitly or explicitly reached for how groups are making decisions and then that is that interesting question where well i mean again groups don't have perceptions cognitions and actions they're just composed of people who do okay fine but then people don't right it's just the cells and that's the um reductionist slippery slope and then one can also um take the train in the other direction the holist slippery slope well individuals be they cells or organisms or groups they're always in an ecological context and so they actually can only be said to be a bit player in their decision making because they're part of these extended enacted networks of action and so it's like we know there's a train that slides us down the reductionist path and we know there's a train that slides us down the holism path and so much work beyond active inference has observed that relationship or tension or dialectic and again active inference is like maybe in that metaphor like the train conductor helping us do something useful and move things back and forth and have fun and hang out on the train while we're in a qualified way and hopefully in a principled and rigorous and accessible way modeling these different scales of the system yeah the uh policy and politics and group decision making that makes sense now it's reminding me of a conversation i had with um professor of politics called laura cram from edinburgh university so her day job is to um is academic and to look at uh uh politics and decision decision making within politics but um outside of that she teaches she's a yoga teacher and um has created a type of yoga called interception yoga or that's what she brands it as um and she cited a very interesting study

where it was looking at um decision making in politics and you know um as the person as a person enters a room into a meeting uh depending on what the what flags are placed in front of them that they see from different countries will influence the decisions they make in the meeting so you know it's uh the this idea of the symbolism of what's presented in front of us influencing the actions we take the decisions or the actions we take and that's just from a flag you know and fascinating stuff mind-boggling stuff that makes me think about pulling or pulling when there's an assessment of sentiment for example within an organization or within a country that is kind of like interception at that group level there's the the palpation that's like the customs are on the border on the interface and then there's the interoception which is like the sensing within and so as i've listened to your work and seeing this area of interoception develop it reminds me of sometimes that external focus that perception cognition and action have had now i think we're seeing new ways of balancing not prioritizing or ranking they're just complementary processes of x tarot and interoception and wouldn't it would be awesome to have unifying ways of talking about them and modeling them so this didn't have to be like this topping of food versus this topping or like sports teams there are not team exteriorception and team interception there's the organismal mystery and the territory then there's how we model it our mapping and sense making process which is participatory and collaborative and built on the shoulders of giants and so on and so to be able to again respect have usefulness in that space brings us back to active inferences focus on both the epistemic and the pragmatic value and how in moment to moment those shift you're doing something useful and then all of a sudden you need to go do an action that's useful because it helps you reveal something an epistemic action and so just again having fluidity to talk about those two sides of the interoception exteroception coin and the two sides of the coin with epistemic and pragmatic two sides of the coin with perception action it's kind of this minimum of two idea that pervades a lot of the discussion that gives us flexibility but also the ability to like drill in and do useful things fascinating uh do you daniel do you have any personal experiences or examples to share of how uh viewing the world through the lens of active inference has influenced your own life changed your behavior or allowed you to see things differently or situations differently than you did before if you can remember life before active when i was a young forager well there's probably many expressible and ineffable ways to address that and i think of it as a show not tell but the example that comes to mind is again related to my ant research as i was just beginning to study this area so on one hand i was reading and learning not doing anything useful from the point of view of perhaps some but i was learning because i

was continually being surrounded by novelty and new information and what was very exciting and just meeting my curiosity with this endless stream of papers and formalisms and then on the other hand i was there in the field with the stopwatch and the ants and so i thought how can there be a synthesis of the ways that the ants research community has modeled the multi-scale system that are the ants and this kind of abstracts notions that i'm learning in active inference and i acted i reached out to authors and i started sketching different models and moved on to a path that resulted for example in 2021 in an active infra-ants model and so that was something where it wasn't on my radar when i started that process of action and curiosity in tandem and so i think that active inference through bigger longer time scale phenomena like just described like adjusting one's research agenda or regime of attention as we might say towards applying active inference to a phenomena that i was already interested in but then seeing how that plays at a much shorter time scale too it's just fun to weave together the concepts and find the patterns across different systems and phenomena yeah i got a sense of the the looping there so you went from curiosity to action reaching out to people and that's that loop has kind of snowballed i suppose and brought you where you are today wonderful so i first became aware of the term active inference after reading it in a paper by seth and tristan on the topic which this channel is all about which is interception and the paper was called active interceptive inference so they they jammed intrceptive in between uh active in inference and the emotional brain and it um so it speaks to the heart of innocence channel the way i understand it is that um what we what we can see is a model where signals from our body so these feelings that we um sense from inside our body can be used to update our predictions about the state of the body so you know i'm now hungry because my stomach's rumbling for example or um and we could just leave it there or we could actually lead to an action that helps us return to helps to keep stay alive basically um return to homeostasis so this is you know my um uh blood vessels dilate or contract and that helps me move towards food for example so we carry on like if i can give it on the temperature example and how active inference deals with homeostasis that return to a normal temperature range or ph or blood sugar range but also how active inference deals with allostasis which is sometimes where stress is actively pursued by an organism or they make a decision that's anticipatory like choosing to bring a jacket with the expectation that it will be colder outside so in those kinds of settings which we interpret as cognitive like if a robot or an ant did it we would think that it was thinking because it planned to bring the jacket along and so that's like allostasis or planning because it's the expectation through time of body temperature and actions can be

taken in anticipation about future expectations somebody can wince before there's a physical impact and there's probably so many other examples so just wanted to add that point that a lot of frameworks deal with homeostasis and the return to like a control region and that's very familiar to the areas like cybernetics and control theory but where are those frameworks without this deep generative model concept sometimes fall a little short is in dealing with situations where there's uncertainty about the future but actions are taken now to ameliorate that uncertainty about the future or to bring body temperature into alignment again in the future but not putting on the jacket in the house but preparing a future affordance you just want to bring that up because that temperature example it's both a basic example used in cybernetics and control theory and homeostasis and physiology but just by moving into that example again qualitatively before we layer on any math or any variables we can think about the reality of how temperature is modulated in the world and that opens up a very rich discussion about other cognitive features that would be really important to consider yeah thank you um yeah the term allostasis um is uh less well-known than homeostasis as you're saying that so you know um lisa feldman barrett has come up with the term body budgeting which as i understand it is equivalent to allostasis and it's so it's got that kind of idea of budgeting or forecasting as you say planning into the future and the body bit is about that it's looking after our you know our energy or resource budget or balance um so the unless you as you use that example of temperature and putting on a jumper i was thinking okay that's that's for me personally and then i might build a house to control my temperature environment for my family and then we have weather broadcasters and weather stations which if you're going back to your point about group um decision making you know if uh the weatherman or woman says tomorrow there's gonna be a snowstorm then that changes the decision that has an act people will take action on a kind of city or country or nation wide scale so um would you say that's allostatic control at a group level or active inference at a national level absolutely the weather person provides a high and a low temperature let's just say but another way that they could share those two temperatures would be like the midpoint and then the variance around that midpoint and so that's still sharing two numbers so instead of saying it's going to be between 20 and 30 we could say it's going to be like 25 plausibly plus or minus 5 and it still could be outside with a smaller probability but 95 or 99 of the time so absolutely what they're conveying is their estimate and uncertainty which is kind of like where epistemic humility comes into play we say things like i'm not sure about or i'm quite sure about and again that's reflected in that weather report is it 28 to 32 or is it 10 to 40. so what is the variance and we

always hold space for that mean estimate and the variance estimate and yes the weather report is being utilized as an actionable signal for people who then take actions from their set of affordances based upon the information they're receiving so then people go and buy a jacket if they hear that there's going to be an upcoming string of cold days for example so yes it's a great example about how communication at the group level is happening through information technology and through meteorology but then also there's that biological version of like different cells and different organs kind of like sending weather reports there was a professor at my graduate school craig heller who made this awesome and interesting technology it was like a glove and there was other variants that had a slight negative pressure and then it had cool but not cold water circulating and it turned out that that was able to reduce the body's temperature and that previously many um approaches that people had taking to recovering people from hypothermic or hyperthermic entailed putting cold or hot appropriately on like their thermal sensors and that actually sometimes suppressed the body's actual thermal response because it's like the body's overheated and the danger is the core overheating but then the thermo sensors like on the shoulders would become cooled down so the person would immediately feel relief but it shut down their endogenous vasodilation actual heat dissipation which created a dangerous situation and so this is like a way of not using active inference but of course could be modeled as such and working with the ways that bodily and group systems actually do communicate and make decisions rather than thinking that we can just intervene in this one specific way i love that example and it sort of reminds me of some of the therapeutic approaches for dealing with something like anxiety or phobias where you're wanting to you know stop the overall system from um going about its you know it's normal action perception loop and find an entry point and give it a different way to update its its model or update its action um so that it doesn't turn into a vicious cycle um and you know that that method i suppose which they had a negative pressure glove or something you said i guess that's similar to some of these techniques like you know the wim hof method or something where the the person is training themselves so that their body doesn't um i guess you could say that the set points the uh allostatic set points are being shifted by the practice or the model of how dangerous the world is is being changed um by the practice so the the perception of threat is being altered entities are taking actions that are aligned with the kind of entity they expect and prefer themselves to be and that's something that for sure can be unpacked and explored but perhaps methods of training like the breathing methods that you mentioned in a sense they update i'm the kind of thing that is chill in a cold shower that's me i'm that person who does

take a cold shower who has the discipline to do so and who sees that being enacted not the kind of entity who observes the book being unread on the table about cold showers but doesn't feel the cold i'm the person who is now perceiving taking deep breaths and all that comes along with that from the vegas etc while i'm also getting that cold water and that is where the updating and the development happens not in the hypothesized experience so i think there's some amazing and deep and relatively lesser explored territory with how we frame our own interception and development and um i mean we i'm tempted to sort of go deep in this on the and go into the free energy principle and expecting ourselves to be in a state and but i'll save that maybe for another time um i am aware we're coming up to an hour and this was um due to being our hour-long meeting um do you i wanted to go into some of the terms um that you use with your your colleagues uh but i think we'll maybe we'll leave that um can you if i could give a closing note there just yeah on a historical and a relatively closing issue then we can however much shortly longer you'd like several years ago and in earlier developmental stages of this framework there was i would say less clarity between what the free energy principle was and active inference and so i hope that increasingly it becomes distinguished and that people can find accessible on-ramps onto both but for those who are listening this far in yes it would be awesome to unpack it in the future but active inference is about that action perception loop and free energy sort of meets there and goes in a totally different direction well i don't know if it's totally different but it is a different approach and so like it is a different way that we can go okay so yeah if anyone watching or listening to this is curious about the free energy principle then there's lots to be enjoyed on that topic one one question before we wind up do you have any thoughts about when interceptive active inference could become maladaptive to an organism as i am only an ant doctor i can only speak to that but it makes me think about interoception and i think well how could it become maladaptive in the colony if its internal perception for example of how much food it has becomes maladaptive what if it's continuing to forage and just store up seeds way beyond what's reasonable what if there aren't enough seeds but it isn't foraging because it's not elicited to do so because it's colony level and interoception isn't happening or what if it's blocking up its entryway every time it gets cloudy and that's actually like a false positive rate that's unacceptable but it would be appropriate to act that way sometimes when it does flood in the desert but it's not that way every day and so i think it's a very interesting and important area to understand how contexts and different people in different cognitive systems find themselves in situations where the ball rolls downhill and it all makes sense yet it goes in a way that we don't prefer and so again it both does make sense

why the body and mind and spirit are compensating in a given way but also we may want to be part of that process and move it to a different space yeah so those two examples are really you know nice examples uh about gathering resources like food you know do i need to keep hoarding hoarding hoarding hoarding or not or that's fine to let the cupboards go go empty we'll be fine everything's okay and the weather one you know the oh it's a little bit cloudy batting down the hatches for the next month uh but nothing actually happened or the other other extreme don't believe the weatherman and then i'm sure there's analogies at the organismal and at the social level well thank you so much for your time and i've really enjoyed this conversation where can people well what's on the horizon for the active inference lab over the next few weeks and where can people find out more about your work awesome yes thanks again and i look forward to continuing the interaction in different ways so to learn more about academic lab head over to activeinference.org and you'll find a lot of resources there you can also find our videos and other content and so on as well as find out about your affordances for participating in the lab so that's definitely the best way to find out more some of our next steps just to name a few include discussions with none other than you where we'll be with some other authors of work on osteopathy in the therapeutic alliance and a lot of other fascinating topics and we continue to develop some educational materials as well as work through the textbook there's a lot of learning and applying that people are involved in inactive lab great exciting times and awesome yeah really fun times i've again i'm looking forward to continuing it and one hour is not enough but one year is not enough so onwards we forage onwards we storage thank you so much daniel thank you see you later bye